

3 Analysis

3.1 Data Overview

3.1.1 Expected:

- fig/text: Data coverage and share of “in operation” deals in the total deals.
- xxxx unique investors identified, around xxxx% of investors registered in tax havens.
- fig: Temporal distribution of land deals

3.1.2 Data availability

Data availability in `data`

Field	Rows	NonMissing	Missing_Count	No data (%)
initiation_year	5343	4661	682	12.8%
country_iso_code	5343	5247	96	1.8%
world_region	5343	5260	83	1.6%
target_country_code_alpha2	5343	5319	24	0.4%
current_intention_of_investment	5343	5323	20	0.4%
investment_type	5343	5323	20	0.4%

Data availability in `desag`

Field	Rows	NonMissing	Missing_Count	No data (%)
initiation_year	25176	21843	3333	13.2%
country_iso2	25176	21894	3282	13.0%
current_implementation_status	25176	22282	2894	11.5%
target_country_code_alpha2	25176	25145	31	0.1%
role	25176	25176	0	0.0%
is_tax_haven	25176	25176	0	0.0%

Value distribution of `current\_implementation\_status` in dataset `data`

Value	Count	Total	Share
In operation	3446	5343	64.5%
NA	700	5343	13.1%

Value	Count	Total	Share
Startup phase	584	5343	10.9%
Project abandoned	345	5343	6.5%
Project not started	268	5343	5.0%

Value distribution of `world\_region` in dataset `data`

Value	Count	Total	Share
Far East and Central Asia	1644	5343	30.8%
Africa	977	5343	18.3%
South and Central America	889	5343	16.6%
Eastern Europe	620	5343	11.6%
Western Europe	462	5343	8.6%
North America	454	5343	8.5%
Middle East	134	5343	2.5%
NA	83	5343	1.6%
Oceania	80	5343	1.5%

Value distribution of `is\_tax\_haven` in dataset `desag`

Value	Count	Total	Share
FALSE	22536	25176	89.5%
TRUE	2640	25176	10.5%

Value distribution of `role` in dataset `desag`

Value	Count	Total	Share
Shareholder	16156	25176	64.2%
Operating Company	3699	25176	14.7%
GUO	2740	25176	10.9%
DUO	987	25176	3.9%
CSH	915	25176	3.6%
ISH	679	25176	2.7%

Value distribution of `from\_tax\_haven` in dataset `investors`

Value	Count	Total	Share
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Value	Count	Total	Share
FALSE	11507	12405	92.8%
TRUE	898	12405	7.2%

3.1.3 How many deals involve tax havens?

Value distribution of `any\_tax\_haven` in dataset `deal\_taxhaven`

Value	Count	Total	Share
FALSE	4025	5343	75.3%
TRUE	1318	5343	24.7%

Value distribution of `num\_tax` in dataset `deal\_taxhaven`

Value	Count	Total	Share
0	4025	5343	75.3%
1	791	5343	14.8%
2	281	5343	5.3%
3	87	5343	1.6%
4	68	5343	1.3%
5	29	5343	0.5%
6	20	5343	0.4%
7	17	5343	0.3%
21	8	5343	0.1%
8	6	5343	0.1%
13	6	5343	0.1%
12	2	5343	0.0%
14	1	5343	0.0%
18	1	5343	0.0%
20	1	5343	0.0%

In operation deals involving tax haven investors

total_in_operation	tax_haven_deals	share
3446	946	27.5%

To quantify the involvement of tax-haven-registered investors, we aggregated the investor-level information in *desag_data* to the deal level. **A deal is considered to involve a tax haven if at least one investor within its ownership chain is registered in a jurisdiction listed as a tax haven.**

The results show that **1318** deals (representing **24.7%** of all deals) involve at least one tax-haven investor. This indicates that tax-haven jurisdictions play a non-negligible role in land investment structures, either through shareholders, controlling shareholders, or ultimate owners.

When considering only deals currently *in operation*, **946** deals out of **3446** are associated with tax-haven investors, corresponding to **27.5%**.

3.1.4 How many unique investors involve tax havens?

Value distribution of `is\_tax\_haven\_investor` in dataset `investor\_taxhaven`

Value	Count	Total Share
FALSE	11507	12405 92.8%
TRUE	898	12405 7.2%

Across the entire database, we identified **12,405 unique investors** involved in at least one land deal. Among them, **898 investors** are registered in jurisdictions classified as tax havens, representing **7.2%** of all unique investors.

Value distribution of `roles` in dataset `taxhaven\_roles`

Value	Count	Total Share
Shareholder	366	942 38.9%
GUO	154	942 16.3%
CSH	144	942 15.3%
Operating Company	138	942 14.6%
ISH	102	942 10.8%
DUO	38	942 4.0%

3.1.5 How does the number of land deals changed over time?

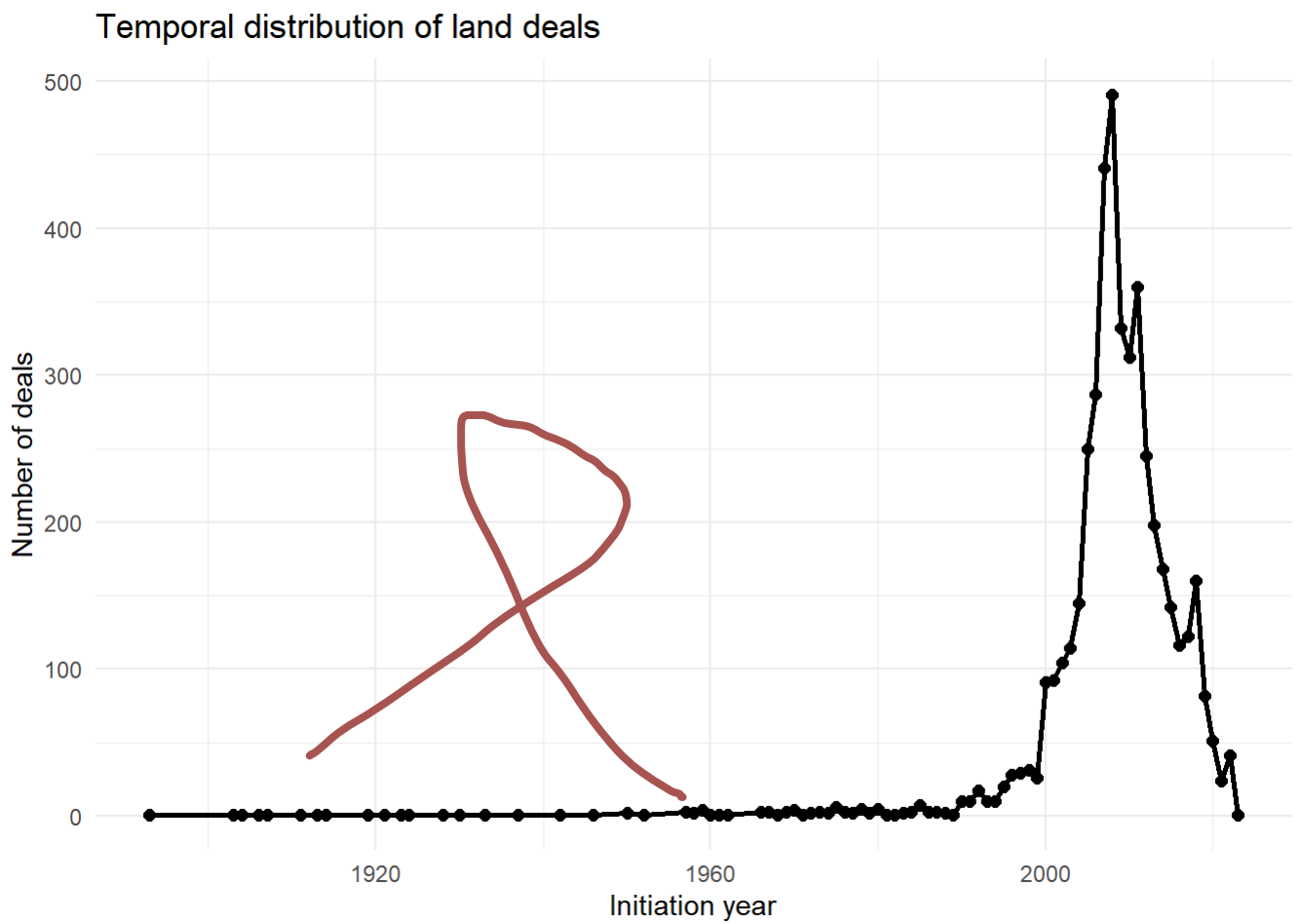


Figure 3.1: Temporal distribution of total land deals

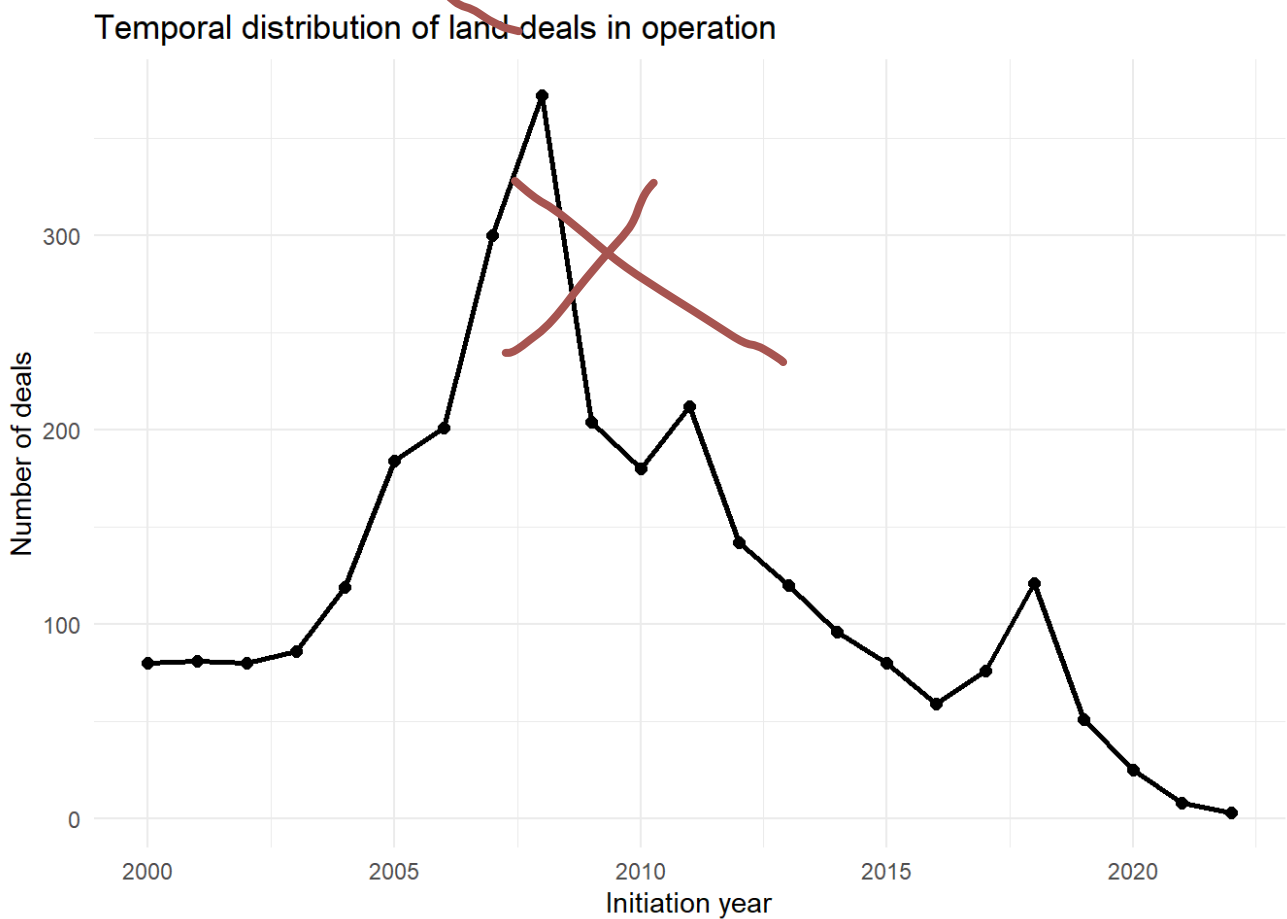


Figure 3.2: Temporal distribution of land deals in operation

In-operation deals since 2000: Total vs Tax-haven involved

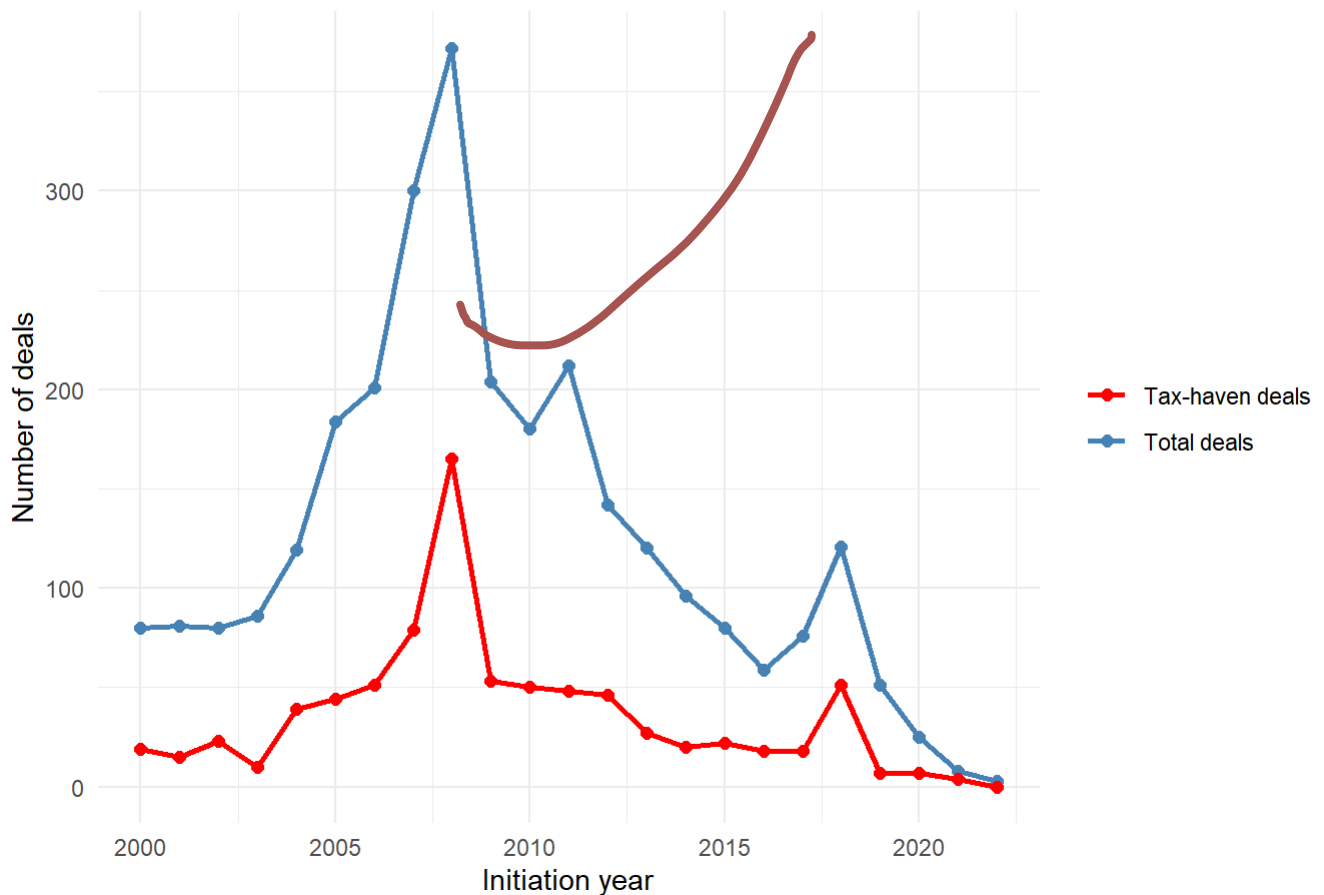


Figure 3.3: Temporal distribution of land deals in operation

3.2 Distribution Patterns

1. bar: Share of tax haven vs. non-tax haven investments
2. line: Evolution of tax haven investments over time -> Analysis of whether the use of tax havens has increased over time
3. bar: Top 10 tax haven investor countries -> Identify the primary source countries of tax havens
4. bar: Top 10 target countries for tax haven investors -> Identify the principal countries of investment
5. bar: Tax haven investments by sector -> Comparing the proportion of tax havens across different investment sectors
6. Map: Global investment flows from tax havens to target countries -> Visualising the global investment pathways of tax havens

3.2.1 Evolution of tax haven investments over time

Temporal distribution of land deals by tax haven involvement

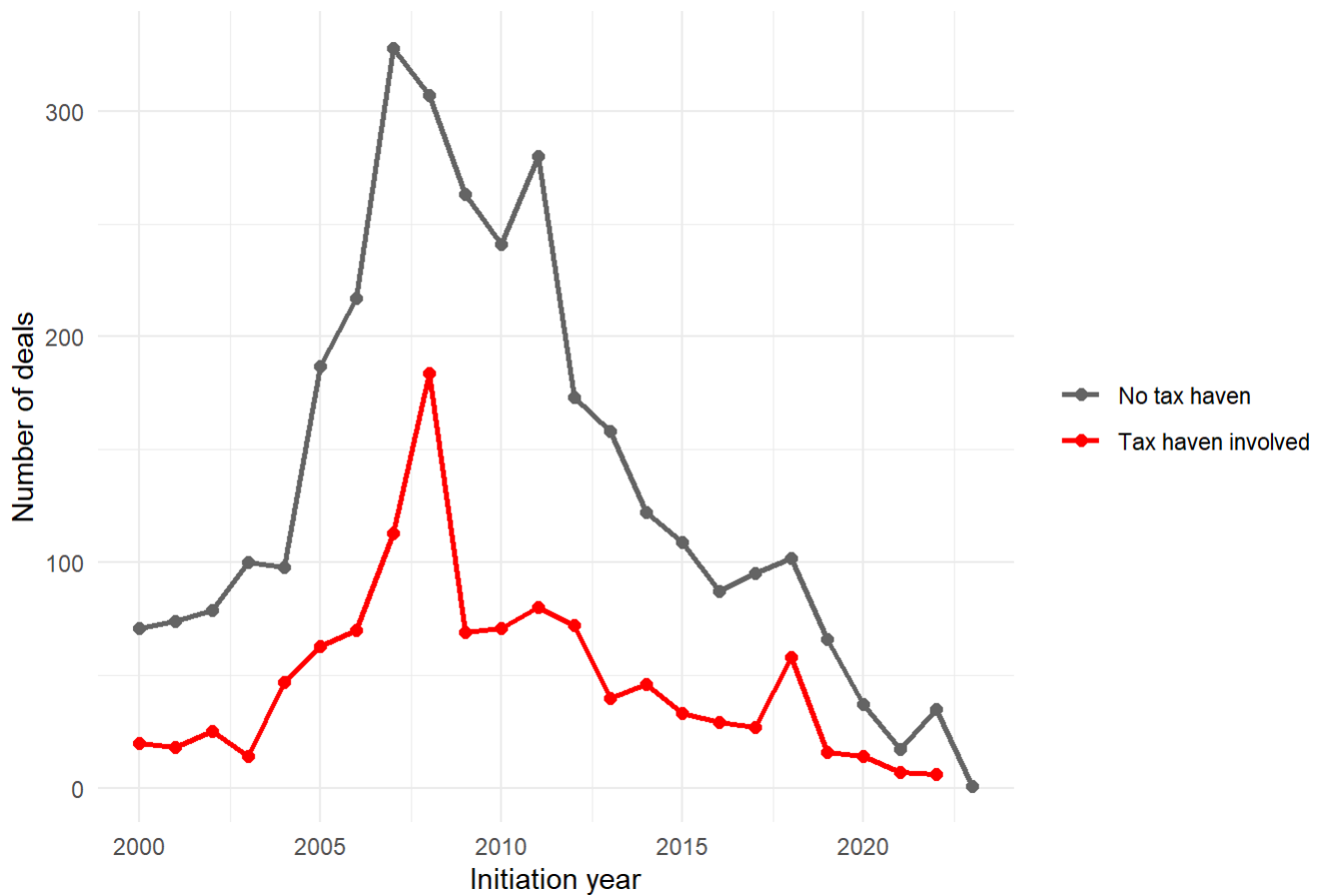


Figure 3.4: Temporal distribution of land deals in operation

3.2.2 Top 10 Tax Haven Investor Countries

3.2.2.1 Which tax haven countries exert the strongest influence on the global land investment network?

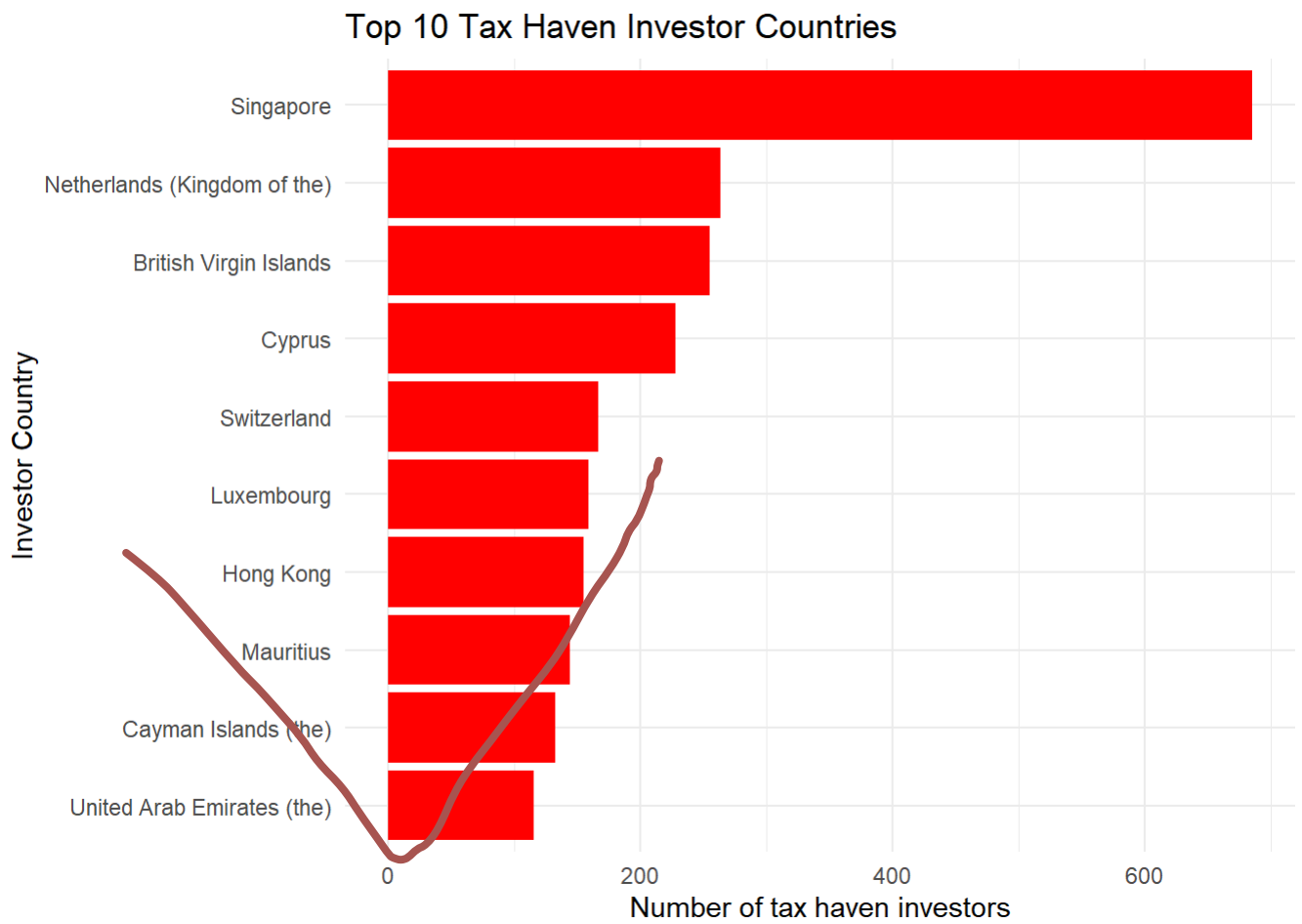


Figure 3.5: Top 10 Tax Haven Investor Countries(appearances)

3.2.2.2 Which countries provide the largest number of offshore entities?

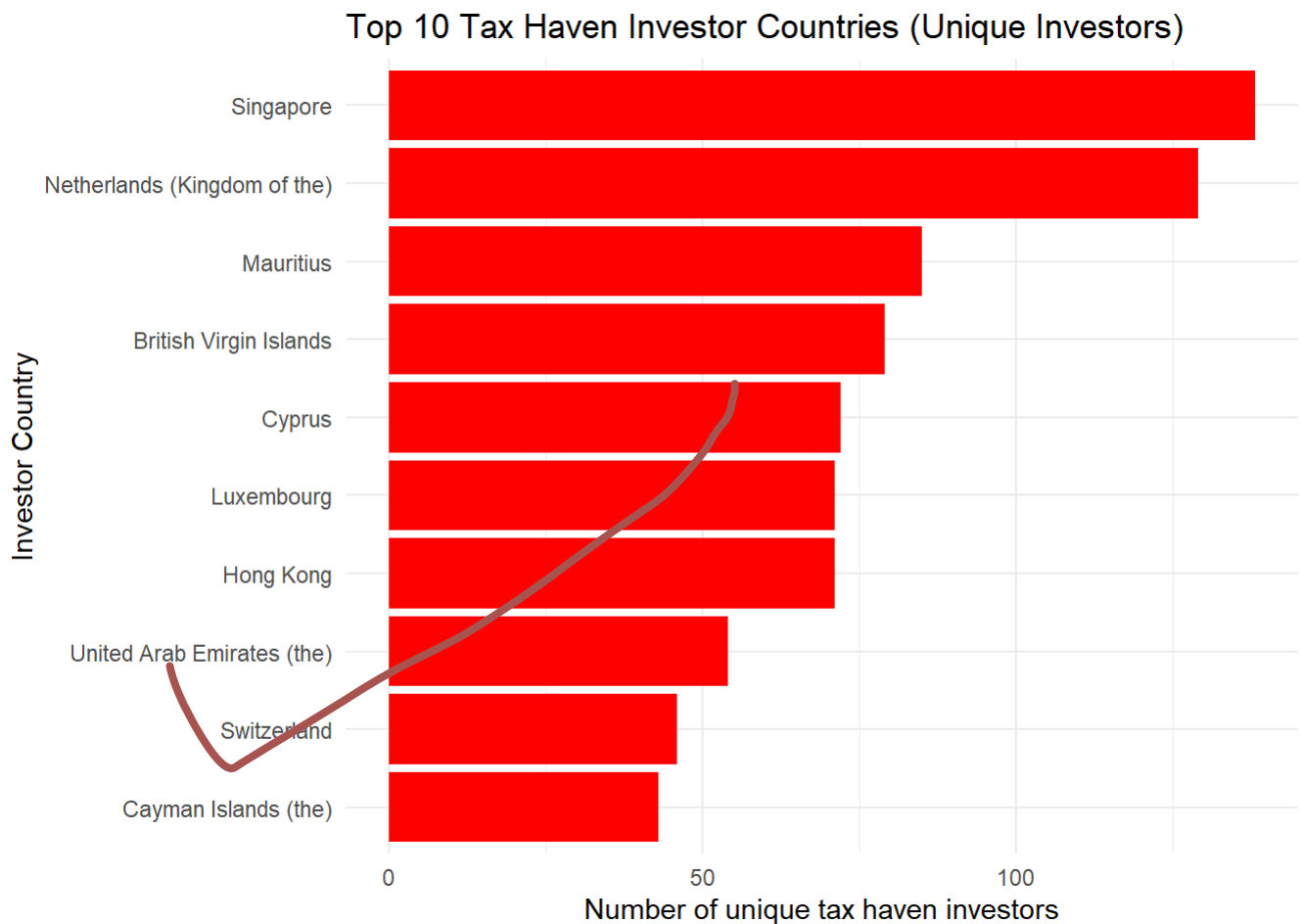


Figure 3.6: Top 10 Tax Haven Investor Countries(Unique Investors)

- **Singapore is dominant in both frequency and breadth**, meaning it is both a *major offshore incorporation hub* **and** home to *highly active entities repeatedly participating in deals*.
- **Some jurisdictions (e.g., British Virgin Islands, Cyprus) show high appearance counts relative to the number of unique investors**, suggesting a **small number of extremely active offshore structures**.
- **Other jurisdictions (e.g., Netherlands, Mauritius) host many distinct offshore investors**, but each investor tends to participate in fewer deals.
→ This pattern reflects **distributed incorporation** rather than intensive participation.
- The comparison reveals **two different modes of offshore involvement** in land investments: **Concentrated influence** (few investors, highly active — e.g., Cyprus, BVI); **Broad-based incorporation** (many investors, moderate activity — e.g., the Netherlands, Mauritius).

3.2.3 Top 10 target countries for tax haven investors

3.2.3.1 Which countries attract the most investment from tax haven investors?

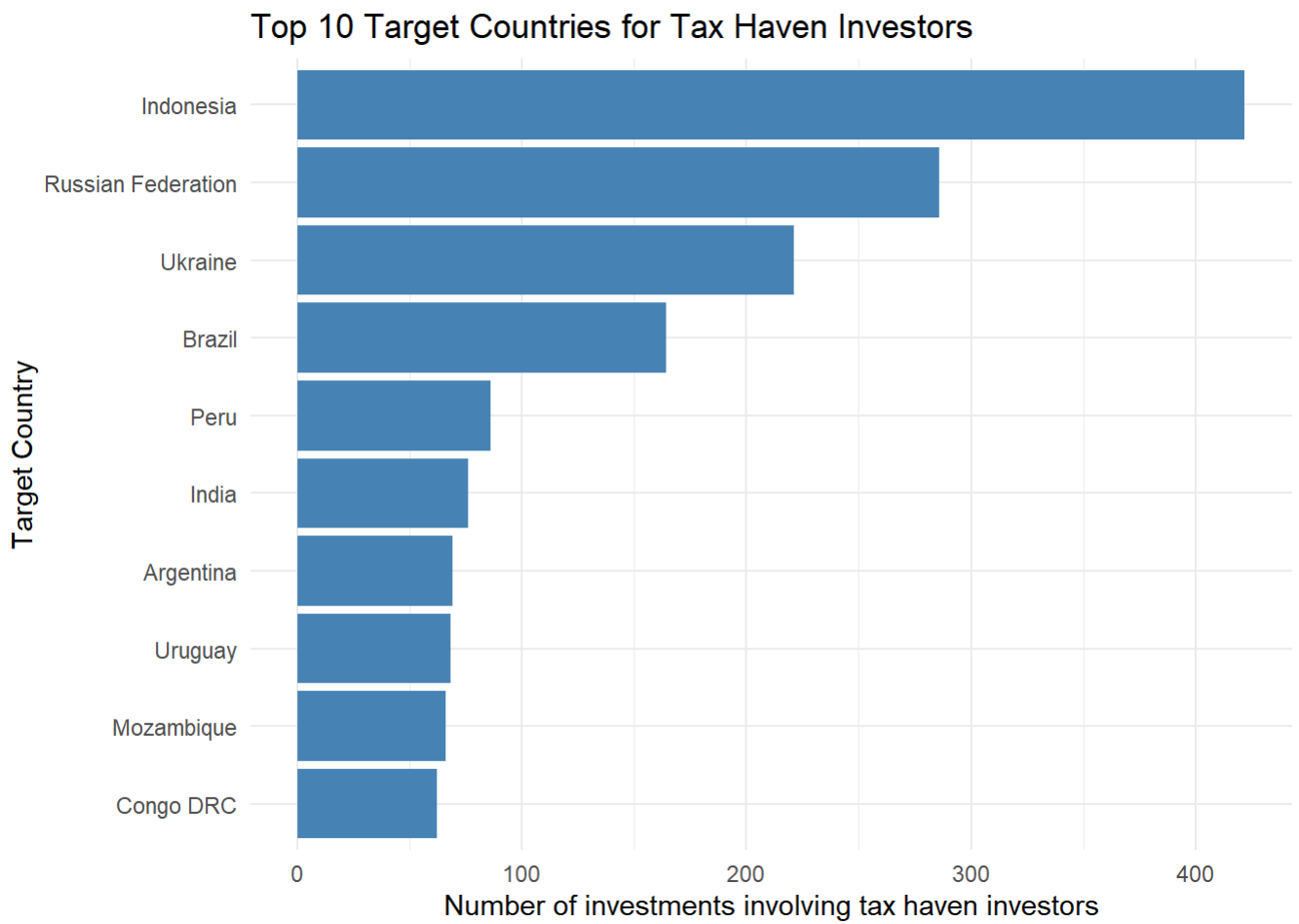


Figure 3.7: Top 10 Target Countries for Tax Haven Investors

3.2.3.2 Which countries' land transactions are most affected by tax havens?

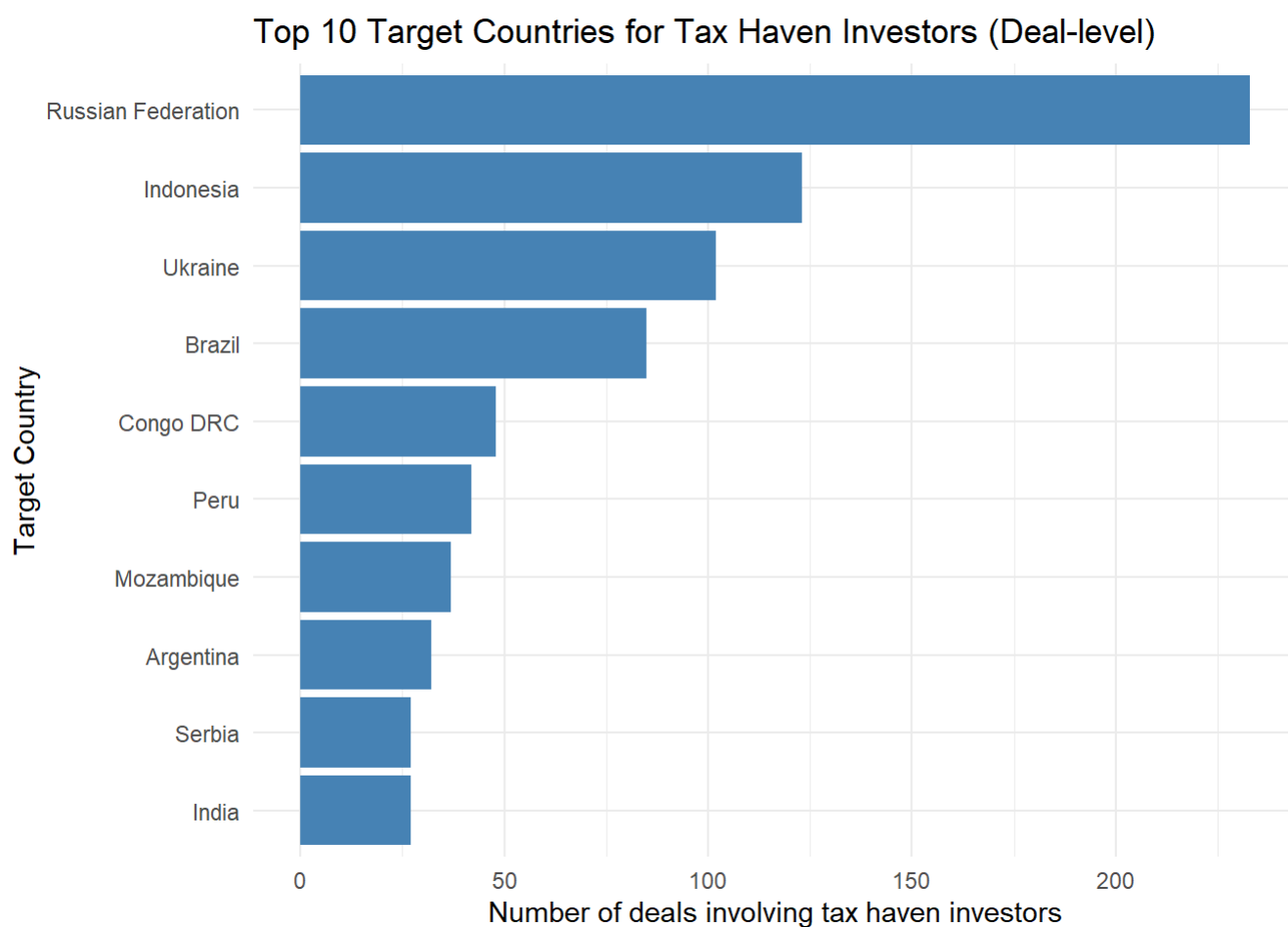


Figure 3.8: Top 10 Target Countries for Tax Haven Investors (Deal-level)

3.2.4 Tax haven investments by sector

Value	Count	Total	Share
Agriculture	11418	25176	45.4%
Energy	4994	25176	19.8%
Mining	4085	25176	16.2%
Forestry	2397	25176	9.5%
Industry	1121	25176	4.5%
Conservation and Carbon	664	25176	2.6%
Other	268	25176	1.1%
Tourism	79	25176	0.3%
Land speculation	77	25176	0.3%
NA	73	25176	0.3%

Value distribution of
`intention_investment_cat` in dataset `desag`

Figure 3.9: Tax Haven Investments by Sector (Share)

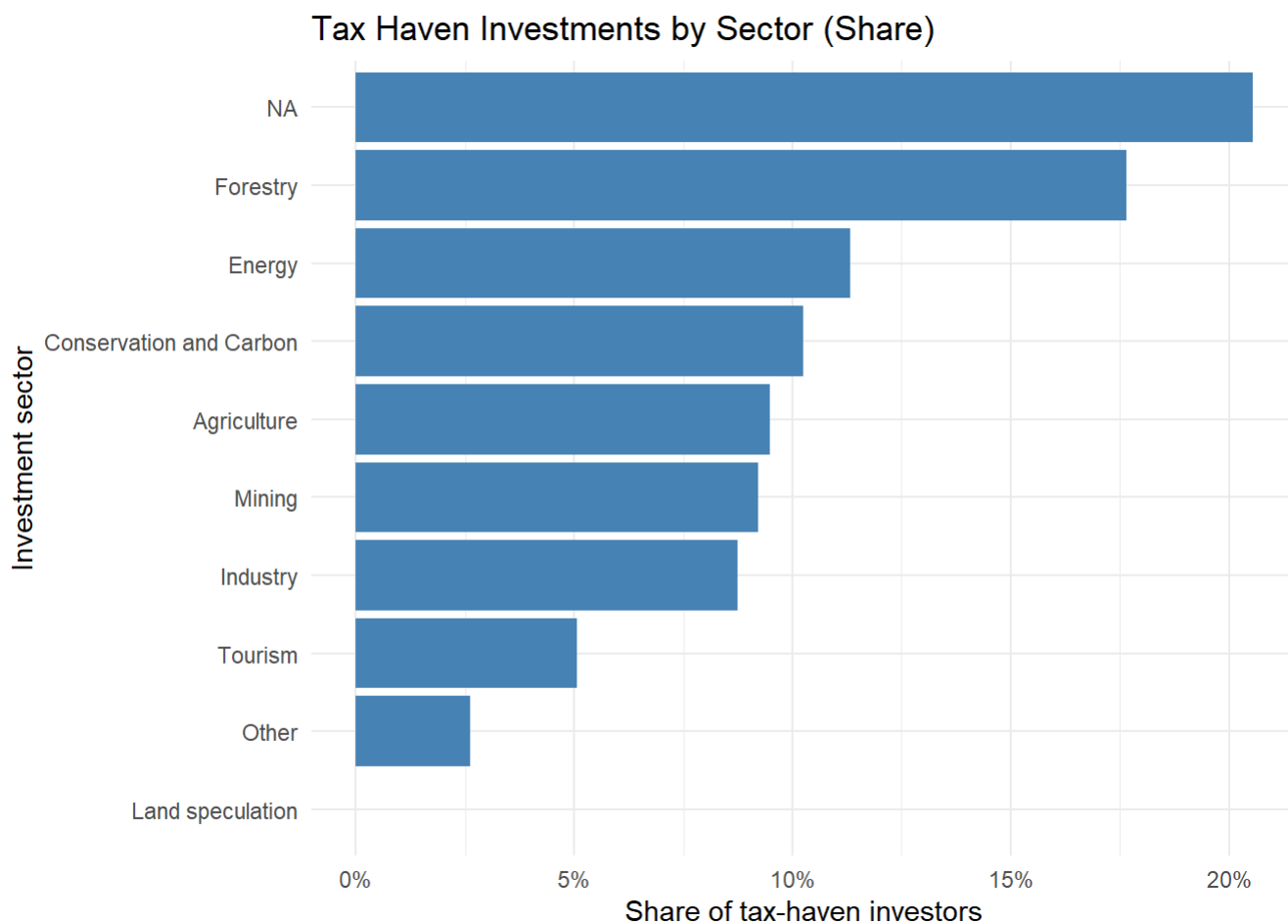


Figure 3.10: Tax Haven Investments by Sector (Share)

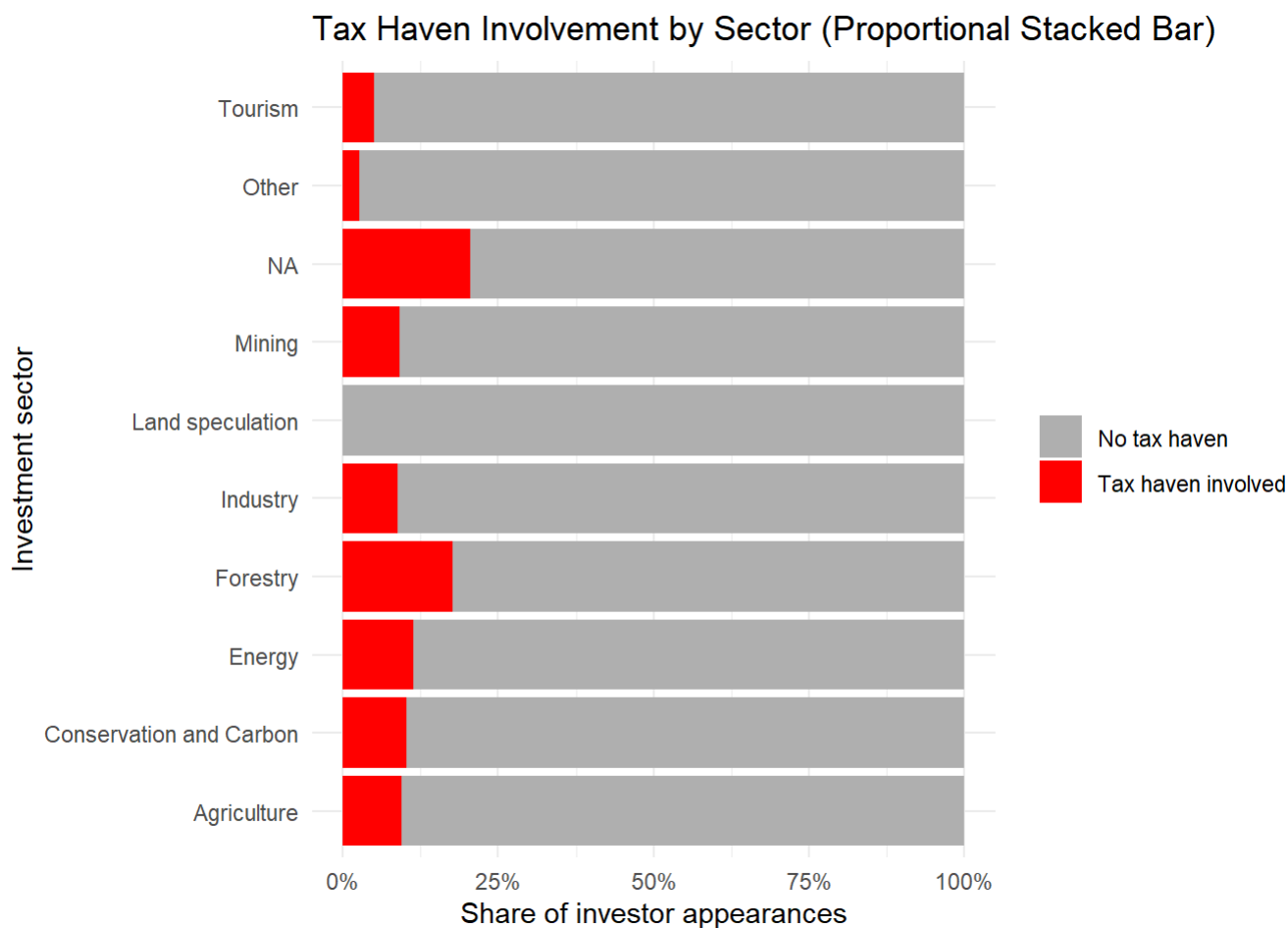


Figure 3.11: Tax Haven Involvement by Sector (Proportional Stacked Bar)

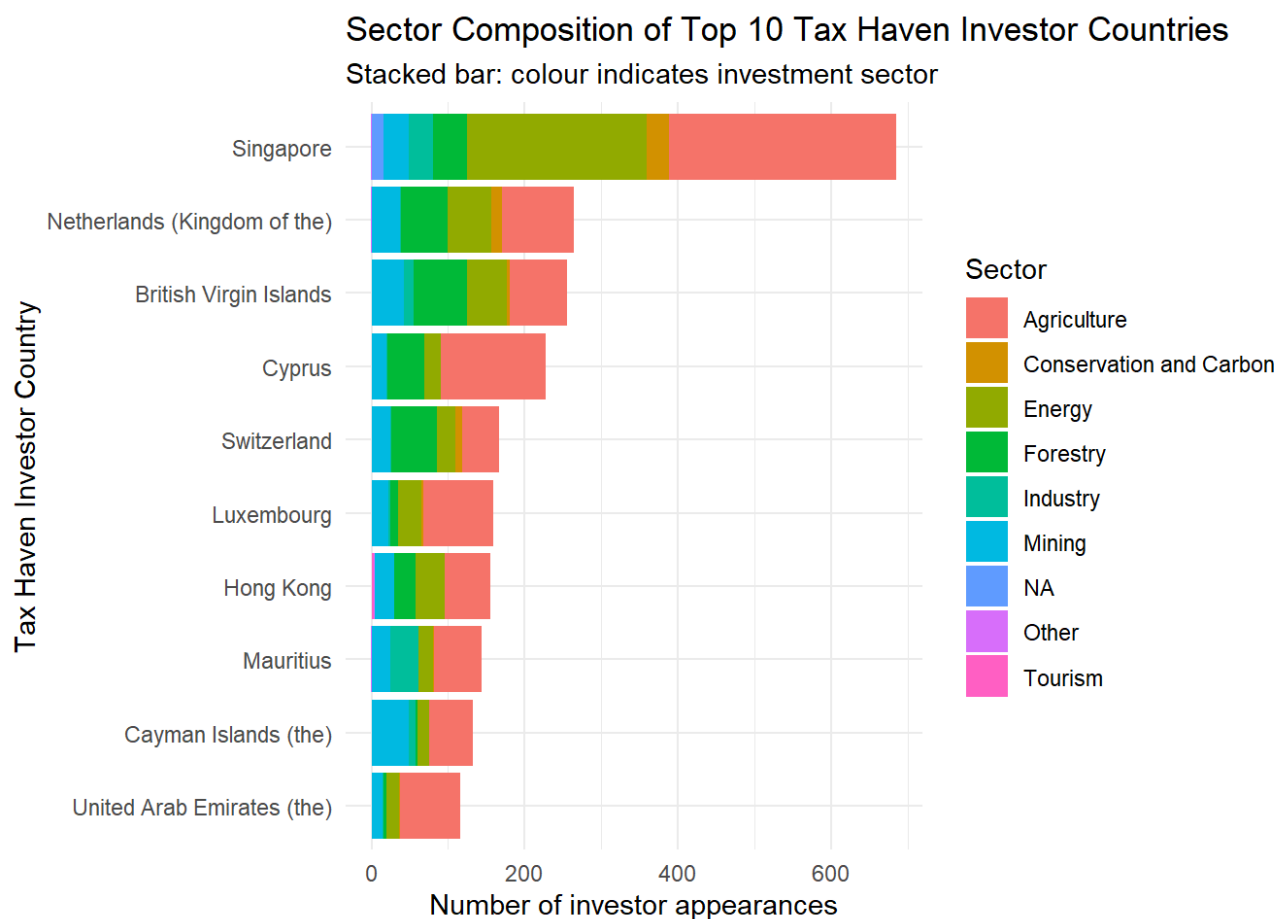
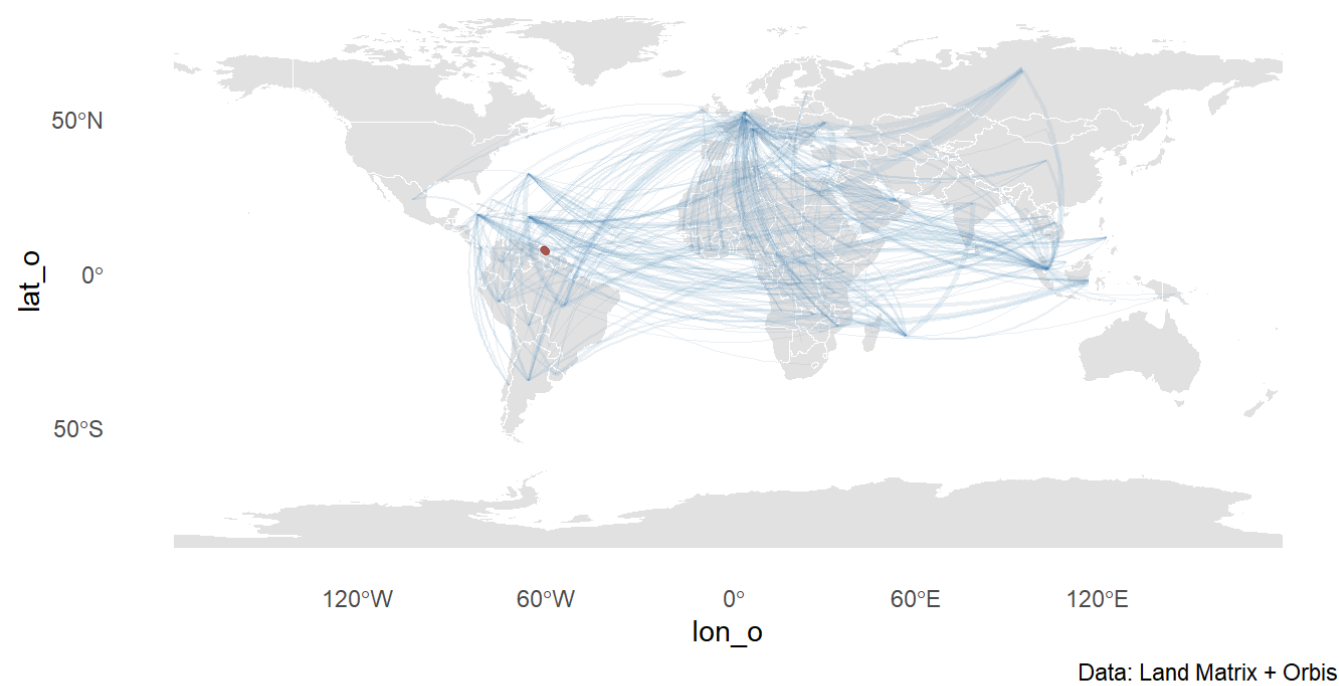


Figure 3.12: Sector Composition of Top 10 Tax Haven Investor Countries

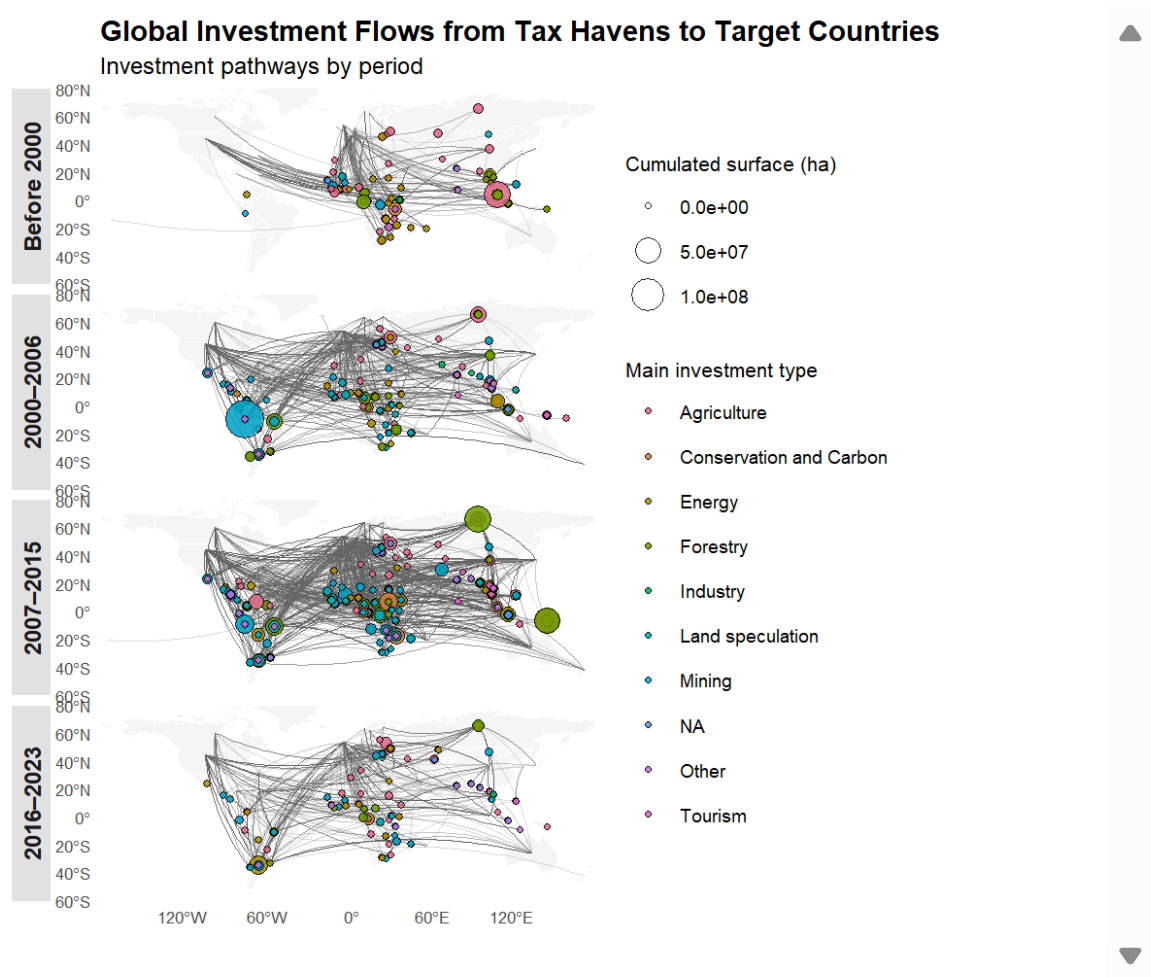
3.2.5 Map [↗](#)

Global Investment Flows from Tax Havens to Target Countries

Each curved line represents investment involvement from a tax haven investor country



{#fig-Global Investment Flows width=672}



3.3 Network Analysis

- Network graph
 - highlighting tax haven nodes
 - Table of top 10 countries by centrality (degree, betweenness)
 - evaluate the bridging role of tax havens;
1. network map: Country-to-country investment network -> Illustrating the global investment structure and the locations of tax haven nodes
 2. table: Top 10 countries by betweenness centrality -> Identifying which tax havens serve as key bridge nodes is crucial.
 3. network map: Regional clusters of tax havens -> Demonstrating the clustering characteristics of Caribbean, European and Asian tax havens
 4. line: Network density evolution of tax haven participation -> Monitor network connectivity trends over time

3.4 Role-Level Analysis

- Examine how tax haven entities appear across **investment roles**:
 - Operating Company (OC)
 - Shareholder (SH)
 - Controlling Shareholder (CSH)
 - Global Ultimate Owner (GUO)
- Summary table: Number of tax haven entities by role. -> Concentration of tax havens across OC, SH, CSH, and GUO tiers
- Sankey diagram: Country → Role → Target country. -> Illustrating investment pathways and the role of tax havens within them

3.5 Impact Assessment
